PRODUCT BROCHURE

GNSS/INS SMART RECEIVER

High-precision positioning and attitude for demanding outdoor operations

Continuous position and attitude information for agricultural machinery, construction equipment, autonomous systems and mapping platforms - from guidance to trajectory acquisition.



FROM FIELDS AND JOBSITES TO ROADS AND WATER

A dependable positioning, heading and attitude foundation for mobile equipment

APPLICATIONS

MULTI-INDUSTRY POSITIONING

01

Precision Agriculture

Auto-steering, implement guidance and path control

02

Construction Machinery

Guidance for excavators, dozers and graders

03

Autonomous Vehicles

Inspection robots, robotic mowers and outdoor vehicles

04

Marine & USV

Heading, surface survey, dredging and marine operations

05

Surveying & Mapping

Trajectory acquisition, LiDAR georeferencing and mobile mapping

TECHNICAL DATA

TECHNICAL SPECIFICATIONS

GNSS / INS SMART RECEIVER



SYSTEM PLATFORM

Product	GNSS/INS smart receiver
GNSS	Unicore UM982
IMU	YJ614E (exact variant TBC)
Core module	Wuhan Vanxoak RK3506 series
Processor	3× Cortex-A7 + 1× Cortex-M0 Up to 1.5 GHz*
OS	Linux
RAM	128 MB
Storage	128 MB (media type TBC)

COMMUNICATION & I/O

Serial	2× RS232
CAN	2× CAN
4G	Cat.1
Wi-Fi	Wi-Fi 6, 2.4 GHz 802.11 b/g/n/ax*
Bluetooth	BLE 5.2*
Connectors	Pinout and baud rate TBC
CAN protocol	CAN 2.0B / CAN FD TBC

RECEIVER MECHANICAL & POWER

Input voltage	12-28 VDC
Max. current	800 mA
Dimensions	250 × 159.221 × 98.563 mm
Weight	TBC
Housing	Material not stated on drawing; TBC
Mounting parts	2× bracket STPA-009-015 2× fixing plate STPA-009-016
Fasteners	Stainless-steel bolts and washers M5×16 parts are 304 stainless steel
Mounting aids	3M tape 100×30×1 mm Insulating pad 8×30×2 mm
Ingress rating	TBC
Operating temp.	-20 °C to +70 °C
Humidity	5%-95% RH, non-condensing
Main connector	776228-1, with brass nut
Antennas / pinout	Quantity and definitions TBC
Certifications	CE / FCC / RoHS etc. TBC

GNSS SIGNALS

Channels	1408, based on NebulasIV
Constellations	BDS / GPS / GLONASS / Galileo / QZSS / SBAS
Master antenna	GPS: L1C/A, L2C, L2P(Y), L5 BDS: B1I, B3I, B1C, B2a, B2b* GLONASS: G1, G2 Galileo: E1, E5a, E5b, E6 QZSS: L1C/A, L2C, L5, L6 SBAS: L1C/A
Slave antenna	GPS: L1C/A, L2C BDS: B1I, B3I GLONASS: G1, G2 Galileo: E1, E5b QZSS: L1C/A, L1C/B, L2C
Modes	On-chip RTK positioning Dual-antenna heading

DATA & CORRECTIONS

Corrections	RTCM V3.X
Output	NMEA 0183 / Unicore
PPP services	B2b-PPP / E6-HAS / QZSS L6E Subject to firmware and service
Network RTK	NTRIP implementation TBC

UM982 MODULE ENVIRONMENT

Operating	-40 °C to +85 °C
Storage	-55 °C to +95 °C
Humidity	95%, non-condensing
Vibration	GJB150.16A-2009 MIL-STD-810F
Shock	GJB150.18A-2009 MIL-STD-810F

POSITIONING & HEADING

Single point RMS	Horizontal 1.5 m Vertical 2.5 m
DGPS RMS	Horizontal 0.4 m Vertical 0.8 m
RTK RMS	Horizontal 0.8 cm + 1 ppm Vertical 1.5 cm + 1 ppm
PPP RMS	Horizontal 5 cm Vertical 10 cm
Heading	0.1° / 1 m baseline
Velocity	0.03 m/s RMS
Timing	20 ns RMS
Cold start	< 30 s
Initialization	< 5 s typical
Init. reliability	> 99.9%
Update rate	20 Hz positioning & heading

YJ614E IMU (614E-B REFERENCE)

Reference	FSS-IMU614E-B Exact product variant TBC
Sensors	6-DOF MEMS 3-axis gyro + 3-axis accelerometer
Attitude output	High-precision attitude angles
Gyro range	±500 °/s
Gyro bias instability	2.5 °/hr
Angular random walk	X/Y: 0.4, Z: 0.3 °/√hr
Gyro thermal bias change	0.03 °/s -40 °C to +85 °C
Accel range	±6 g
Accel bias instability	25 μg
Velocity random walk	0.05 m/s/√hr
Accel thermal bias change	XY: 0.5 mg, Z: 2.0 mg
Output rate	Up to 1000 Hz
Measurement delay	7 ms
Interfaces	Serial / I²C / SPI / CAN
Size / weight	17 × 14.7 × 3.2 mm / 2 g
Shock / vibration	2000 g / 10 g
Environment	-40 °C to +85 °C compensation 100% magnetic shielding

RECEIVER-LEVEL DATA REQUIRED

Still required: exact IMU suffix (B/C/G/Q/U), receiver weight, housing material, ingress rating, antenna/pin definitions and certifications.

Dimensions, electrical ratings, temperature, humidity and mounting materials follow the supplied DG-NJB-030 assembly drawing. GNSS data follows UM982 specifications; IMU data follows the Forsense FSS-IMU614E-B manual.